Errata and model clarifications — *Self-Spectre, Write-Execute and the Hidden State*

Errata to G. Morse, Tatra Mt. Math. Publ. **73** (2019), 131–144, DOI 10.2478/tmmp-2019-0010

Page numbers refer to the published version. Items marked *Lean* are proved in the accompanying formalization (`lean/`); items marked *listing* are checked against the assembled MASM objects (evidence, not proof); items marked *editorial* are corrections of the text.

Corrections to the text

W01 (p. 141, Section 4, paragraph after Figure 7; Lean) Replace "A full compiler can be written and Turing-completeness could be readily proven from the model as already presented." by "A full compiler can be written. Turing-completeness holds for the idealized machine with unbounded code memory initialized with a repeating blank-cell template, in which each cell encodes its symbol in the displacement of one jump and write-only handlers rewrite it. For a finite image the simulation is exact while the head stays inside the image, and the machine faults when it leaves.". *Lean:* `Universal.universal_halts_iff`, `Universal.never_faults`, `Universal.bounded_halts`, `Universal.bounded_faults`.

W02 (p. 139, Section 4, paragraph beginning 'The practical details'; listing) Replace "First of all, no registers are used in the instructions, and only jumps and data movement instructions are used." by "No Write-Execute procedure uses the stack: each executed instruction is an immediate store into an instruction field, a direct jump, or a no-operation. Registers and returns occur only in the C-calling-convention entry points, which patch the argument fields and return the result.". *Lean:* `WE.step_eq_effect`, `WE.effect_congr`, `WE.store_from_window`.

W03 (p. 139, Section 4, primitives list; Lean) Replace "division with a 64-bit divisor, and 32-bit dividend, quotient and remainder" by "division of a 64-bit dividend by a 32-bit divisor, with a 32-bit remainder and the quotient modulo 2^32". *Lean:* `Arith.doDivision_spec`.

W04 (p. 136, Section 3, RSA paragraph; Lean) Replace "It also requires a 64-bit divisor divided by a 32-bit dividend with a 32-bit remainder. Although the quotient cannot overflow 32-bits of output, the remainder potentially can." by "It also requires a 64-bit dividend divided by a 32-bit divisor with a 32-bit remainder. The remainder always fits in 32 bits, but the quotient overflows 32 bits unless the high half of the dividend is less than the divisor, which holds when both factors are first reduced modulo the divisor.". *Lean:* `Arith.div_lt_word_iff`, `Arith.mulmodFixed_spec`.

W05 (p. 136, Section 3, RSA paragraph; Lean) Replace "an efficient implementation which requires log2 E + count_bits(E) modular multiplications" by "an efficient implementation which requires floor(log2 E) + 1 + count_bits(E) modular multiplications for E >= 1". *Lean:* `Arith.mulCount_log2`.

W06 (p. 137, Section 3, discussion of Tables 1-2; editorial) Replace "93+5*44" by "93+6*44".

W07 (p. 131, Title footnote, Mathematics Subject Classification; editorial) Replace "62K05" by "94A60, 68M07".

W08 (p. 134, Section 2, fourth paragraph; editorial) Replace "which flushes the cache 4 bytes at a time" by "which performs a 16-byte non-temporal store that bypasses the cache and thereby evicts the line".

W09 (p. 133, Section 2, Meltdown paragraph; editorial) Replace "256 local 512-byte pages" by "256 local 4096-byte pages (the Spectre proof of concept uses a 512-byte stride)".

W10 (p. 140, Figure 5, DEFINENOT (and the identical DEFINENZ); listing) Replace "IF Index GT 0f6h" by "IF Index GT 0f7h". *Lean:* `X86.w10_original_sizes`, `X86.w10_corrected_sizes`, `X86.fig5_x64`, `X86.fig5_corrected`.

Code supplement

The accompanying sources were corrected by minimal diffs (`lean/errata/minimal.diff`):

- **L01** `writeexec.c`: diagnostics for Te1-Te4 print the matching table and the actual result (commit fd49944)
- **L02** `spectreopt.c`: fastmodexp: result = 1 % mod (b^0 mod 1 = 0) (commit fd49944)
- **L03** `spectreopt.c`: mulmod: reduce the high half with % instead of / (commit fd49944)
- **L04** `writeexc.asm`: DEFINENOT and DEFINENZ threshold 0f6h -> 0f7h (commit fd49944)
- **L05** `writeexc.asm`: WriteExecMODEXP32: documented precondition mod >= 2 (RSA moduli); harness excludes mod = 1 for the WE call (commit fd49944)